



S.V. National Institute of Technology, Surat

Department of Electronics Engineering

MID Semester Exam, March-2023

B. Tech. (EC) VI Semester

Communication Networks (EC322)

Date: 15/03/2023 & Time: 02:00 PM to 03:30 PM

Maximum Marks: 30

Duration: 1:30 Hours

Instructions:

1. [CO] indicates course outcome.
2. All the questions are compulsory.
3. Marks distribution for each question is clearly provided in the 'Marks' column.

Q.1	Write the answers in a few words.	Marks	CO
	a Draw the TCP/IP protocol suite. Describe the work of each layer.	1+2	[CO1]
	b Take 5 nodes, connect those in mesh, bus and ring topologies.	3	[CO1]
	c Explain PDR, Throughput and Latency (each component of latency)	3	[CO4]
	d Draw the flow chart of Aloha and explain each step.	3	[CO2]
	e Draw the flow chart of CSMA 1-persistent, non-persistent and p-persistent algorithms.	3	[CO2]
Q.2	Write the short answers to the following Questions.		
	a Write the differences among the passive hub, active hub, 2-layer switch, router and gateway.	3	[CO1]
	b A pure ALOHA network transmits 100-bit frames on a shared channel of 200 kbps. What is the throughput if the system (all stations together) produces a. 1000 frames per second b. 200 frames per second c. 50 frames per second.	3	[CO1]
Q.3	a A network using CSMA/CD has a bandwidth of 10 Mbps. If the maximum propagation time (including the delays in the devices and ignoring the time needed to send a jamming signal) is 51.2 μ s, what is the minimum size of the frame?	3	[CO1]
	b How the IFS can be used to prioritize a node?	2	[CO2]
Q.4	a Assume that, in a Stop-and-Wait ARQ system, the bandwidth of the line is 1 Mbps, and 1 bit takes 20 ms to make a round trip. What is the bandwidth-delay product? If the system data frames are 1000 bits in length, what is the utilization percentage of the link?	2	[CO1]
	b What is the utilization percentage of the link in question 4(a) if we have a protocol that can send up to 15 frames before stopping and worrying about the acknowledgments?	2	[CO1]